MULTI-LAYERED ELECTROSTATIC FILM ACTUATOR

Saku Egawa and Toshiro Higuchi

Institute of Industrial Science
University of Tokyo
7-22-1 Roppongi, Minato-ku, Tokyo 106, Japan

ABSTRACT

There is a strong need for forceful and small size actuators. The authors propose the multi-layered electrostatic film actuator, aiming at realizing high force/weight ratio actuators applicable for large scale machines. It has numerous films with plenty of small electrodes stacked and connected. In order to build this structure, an unique electrostatic actuator named the image charge stepping actuator have been developed. It utilizes image charges induced on high resistor to produce force. It have been fabricated on polymer films and a prototype of the multi-layered actuator with four film actuators have been constructed. The fabricated multi-layered actuator can produce force of 0.4N.

INTRODUCTION

At the present time, mechanical systems whose sizes are about one centimeter to one meter are most often actuated by electromagnetic motors. However, conventional motors are much heavy and large compared with human or other animals' muscles. This brings about a serious problem especially in designing robots because in the robots, their actuators must often carry weight of actuators by themselves. In order to overcome this problem, various new actuators have been developed, such as actuators using shape memory alloy [1], pneumatic muscle [2, 3], and mechanochemical actuators using gels [4].

The electrostatic actuator is a kind of electric motors which has been long studied [5]. Several electrostatic motors whose dimensions and structures are analogous to conventional electromagnetic motors were developed [6, 7], but they could not produce enough force for actual use. This is because they did not take full advantage of electrostatic force. Advantages of electrostatic actuators will be summarized that they suit for miniaturization. They need only thin plane electrodes to produce electric field and do not require large and heavy three dimensional structures such as iron cores and coils in electromagnetic motors. In addition, maximum force per unit volume increases as size of an actuator is reduced, since breakdown field strength of air increases in very small gaps, and interfacial area per volume is enlarged while force is in proportion to the area [8, 9].

Recently, several micro electrostatic motors for micro mechanisms have been designed and developed using silicon micromachining technique [8, 9, 10, 11, 12, 13]. In the field of micro mechanisms, electrostatic force is considered to be advantageous. However, not only in micro mechanisms, but also in ordinary scale machines, electrostatic force can show its talent. The actuator which comprises a large number of small actuator elements and which has large interfacial area [9, 14] can take advantage of miniaturization and at the same time can obtain large force, adding the forces of the element actuators. Theoretically, the total force of such an actuator can be increased to any extent by reducing size of the elements and increasing the number of them, so that high force/volume ratio can be realized. This actuator will be applicable for various machines.

Aiming at realizing forceful and small size actuators, we have been developing the multi-layered electrostatic film actuator, in which numerous films with plenty of small electrodes are stacked and connected [14]. In order to build this structure we have developed an unique actuator named the image charge stepping actuator [14]. In this paper, we describe the concept of the multi-layered actuator, the principle of image charge stepping actuator, its fabrication on polymer films, and stacking of the films. Experimental results are also provided.

MULTI-LAYERED ELECTROSTATIC FILM ACTUATOR

As described above, large force can be produced by an electrostatic actuator with plenty of small elements (electrodes) and large interfacial area. For the material of this actuator, thin fibers or films seem desirable because they allow large surface per volume. Surface area per volume of fibers is larger than that of films but fabrication technique of electrodes on fibers has not been established and its development will be very hard. On the other hand, conventional photo lithography process is applicable to films. Thus we have adopted films, especially polymer films other than silicon substrates which are widely used in recent micro mechanisms, because polymer films are more flexible, less brittle, less expensive, and easier to get large thin films.

When using films, attention should be paid to the fact that films bend and wrinkle easily. However, films will not much be deformed when they are simply pulled straight. Therefore, for film actuators, linear motion is preferable. Considering this, we have adopted a structure shown in Fig. 1, where films are bound into two arrays and they are interleaved. This actuator resembles muscle in the sense that it has interleaved structure and that it produces tensile force. It may be called "*Electrostatic Artificial Muscle.*" Although this actuator looks alike MIPA of Jacobsen et al. [9], there are some differences. They imply the use of silicon or some other hard materials while we use polymer films. Our actuator operates on a principle different from that of MIPA, which is described later.

Effect of reducing size of elements is explained as follows. Electrostatic force acting on an object can be calculated by integrating the Maxwell's stress over a closed surface surrounding the object. Since the Maxwell's stress is in proportion to energy density of electric field E, force of one unit of actuator is estimated as

$$f_u \propto \tfrac{1}{2}\varepsilon E^2 s \propto E^2 l_0^2$$

where ε is permittivity, s is effective interfacial area of the unit actuator, and l_0 is the characteristic dimension. Thus force density per area of the film becomes

$$f_s = f_u / s_u \propto E^2$$

where s_u is area of one unit. This indicates that force produced by a film is definite regardless of size of its elements if strength of field keeps constant. However, thickness of the actuator h can be reduced in proportion to size of elements. Thus the number of actuators which can be stacked in a unit height increases, that is, force density per volume is in inverse proportion to l_0.

$$f_v = f_s / h \propto E^2 / l_0$$

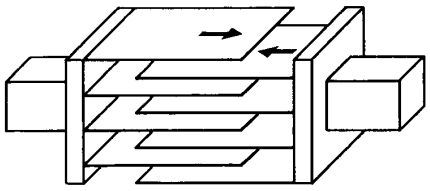


Fig. 1 Schematic Figure of Multi-Layered Film Actuator

(Electrostatic Artificial Muscle)

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Thus, as size of elements is reduced, force of an actuator of a certain volume increases. In addition, since applied voltage V is in proportion to El_0, it can be decreased and the voltage control circuit become easier. Moreover, field E can be increased as already mentioned, obtaining larger force multiplied by square of increase of E. These indicate the advantage of reducing size of elements.

BUILDING MULTI-LAYERED FILM STRUCTURE WITH CONVENTIONAL ACTUATORS

There are several types of electrostatic actuators. Which of them fits best for the multi-layered actuator?

The variable capacitance motor [6, 8, 10, 12] is the most conventional electrostatic actuator. In its linear form, it has parallel striped electrodes on both of two parts which shift relatively to each other. (For simplification, here we call the parts *stator* and *slider*, although it is not absolutely defined which one moves.) Many problems should be solved to fabricate this actuator on polymer films and to stack these film actuators.

(a) It produces normal attractive force besides tangential motive force. To maintain its air gap between the stator and the slider, it requires bearings, whose fabrication on films is difficult. Another way to cancel the normal force is to put the slider between two stators. However, this approach is ineffective for film actuators because it is hard to keep a narrow air gap over large area since the normal force will bend films easily. If the stator and the slider touch each other without lubrication, the normal force will cause so large friction that the slider can no longer move. If the films are hardened by thickening in order to be able to keep the air gap, force density decreases and the advantage of using films disappears. Technique for fabricating bearings on films or some other way to keep the gap and reduce friction should be developed.

(b) Electrodes on the slider and the stator have to be aligned so that they run parallel with each other. Otherwise, the actuator cannot function because its force will be canceled in total since direction of force may vary on places. Alignment of films with long and narrow-pitch electrodes requires very high precision. Moreover, when this actuator is stacked, difficulty increases since films of every actuator have to be aligned respectively. Technique for high precision fabrication and assembly should be developed.

(c) Since films are extensible, they are tend to be imprecise in their shape. Thus displacement between electrodes of the slider and the stator may vary on different places. Therefore, local feedback is required to attain good performance. Fabrication technique of sensors and voltage switching circuits on films should be developed.

The induction motor [6, 7, 13] is another type of the electrostatic actuator. It has a slider made of or coated with slightly conductive material. When traversing electric field produced by the stator is applied on the slider, electric charge is induced on the slider with some lag behind the field. The displacement between the field and the charge causes motive force. Since the induction motor has no electrodes on the slider, it does not require alignment between the stator and the slider nor feedback control. However it has normal attractive force as same as the variable capacitance motor. This causes same problem as (a) above.

There are also a family of actuators classified as *the variable gap electrostatic actuator* [15]. In these actuators, their movable parts are in rolling contact with their stators [9, 11]. Therefore friction in these actuators is small. However, this configuration is probably not applicable to film actuators.

IMAGE CHARGE STEPPING ACTUATOR

Since actuators devised up to the present have problems in constructing the multi-layered film structure, we have developed an unique actuator using high resistor, named *the image charge stepping actuator*, which is free from many of those problems. Fig. 2 shows the basic structure and operating principle of the image charge actuator. Its stator has striped electrodes covered with isolative material. The electrodes are wired so that every three electrodes have same potential. The slider has no electrodes but consists of isolator and high resistor layers. It is driven through the charge induction stage (1 and 2 in Fig. 2) and the actuation stage (3 and 4) as below:

(1) The slider is just put on the stator. The slider has no electric charge at first. Positive and negative voltages are applied on two groups (phases) of electrodes respectively as shown in Fig. 2 (1).

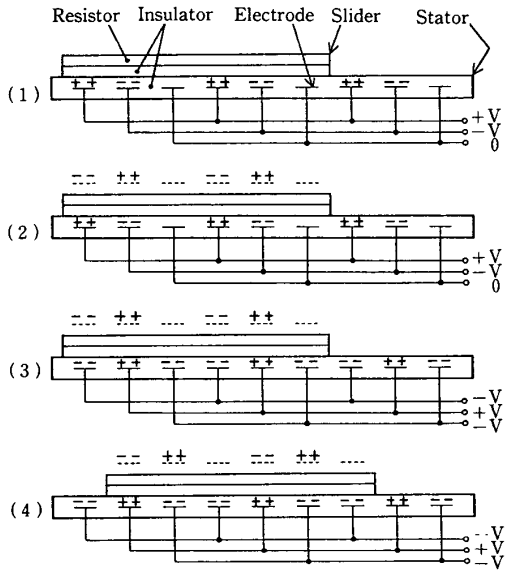


Fig. 2 Structure and Principle of Image Charge Stepping Actuator

(2) Electric field produced by the electrodes causes movement of charges in the resistor which continues until the induced charges neutralize the field in the resistor. In the equivalent state, the induced charges can be indicated by a mirror image of charges on the electrodes, displayed as broken lines in Fig. 2 (2).

(3) After charges are sufficiently induced, the signs of voltages applied on the two groups at the first step are inverted and negative voltage is applied on the third group as shown in Fig. 2 (3). This causes an instant change of charges on the electrodes. However, charges in the resistor cannot move in a short time, obstructed by the high resistance. Thus, charge configuration shown in Fig. 2 (3) appears. Although it requires complex calculation to exactly predict produced force, its direction can be estimated. In this configuration, the negative charge on the third group repels the upper right negative charge in the resistor and attracts the upper left positive charge. Tangential motive force which thrusts the stator to the right is produced as the sum of these two forces. In addition, normal repulsive force is produced because the induced charges on the resistor have same signs as the charges on the electrodes just below. This force reduces friction between the stator and the slider. It helps the slider to move.

(4) The stator halts after it moves about one pitch of electrode. This procedure can be repeated, shifting the voltage pattern applied to the electrodes.

The image charge actuator fits for the multi-layered actuator as described below.

(a) It does not need alignment between films, feedback control, nor high precision in dimensions of electrodes. This is because, instead of having electrodes on the slider, the image charge actuator utilizes induced image charges, which act as a precise copy of the electrodes on the stator. The image charges can also be regarded as short life electrets temporarily made on the slider.

(b) It can function even though it has no bearing and the slider is in contact with the stator because the normal repulsive force reduces friction at the moment when the motive force thrusts the slider. When the slider is stationary, large friction is produced contrarily, which holds the slider tightly. Moreover, since there is no air gap, small separation between the electrodes and the resistor can be easily realized over large area.

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(c) The multi-layered actuator with small number of layers can be constructed simply by stacking and connecting image charge actuators fabricated on films. It requires no complex additional structures since its layers can bear weight of upper layers with the normal repulsive force.

The image charge actuator resembles the induction motor. However, the induction motor uses continuously traversing field and it always has normal attractive force. The image charge actuator changes field discontinuously. It enables the repulsive force to be produced. Moreover, slide of charges in the resistor is smaller than that of the induction motor. Therefore, energy loss in the resistor is smaller.

RESISTIVITY CONTROL

The most difficult problem in making actuators using high resistor is to control its resistivity [13]. In the image charge actuator, it should be selected so that the time for charging and discharging are in the range of about 10 times to 100 times as much as the time for traversing one pitch. If resistivity is too high, it takes too much time for charge induction. If resistivity is too low, the induced charges disappear before the slider moves in the actuation stage.

To calculate the first estimate of the required resistivity, the simplest model of one actuator unit shown in Fig. 3 can be used. The lower plates of the two capacitors represent the two electrodes on which voltage is applied in the induction stage. The third electrode is omitted because it has smaller effect here. Considering dimensions of this structure gives

$$C = \varepsilon \frac{L\, w_C}{d_C}, \quad R = \rho \frac{d_R}{L\, w_R}$$

where L is length of electrodes and ρ is volume resistivity. Thus, time constant of charging and discharging becomes

$$\tau = \tfrac{1}{2} CR = \varepsilon \rho \frac{w_C d_R}{2 d_C w_R}$$

To obtain estimation of exponent of resistivity, we consider dimensions w_C, d_C, w_R, and d_R to be all approximately equal to the characteristic dimension l_0 and ignore the coefficient 1/2. Thus,

$$\tau \simeq \varepsilon \rho$$

The right side is known as relaxation time of free charges in resistor. When very thin coating is used for the resistor layer instead of bulk resistor, surface resistivity σ characterizes it. Then,

$$R = \sigma d_R / L$$

Thus,

$$\tau = \varepsilon \sigma \frac{w_C d_R}{2 d_C} \simeq \varepsilon \sigma l_0$$

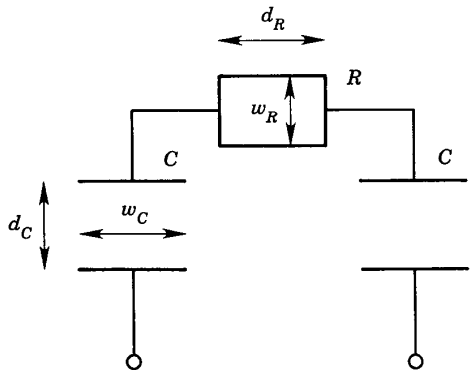


Fig. 3 Simple Model of One Unit of Image Charge Actuator

If we assume, considering the actually fabricated actuator, $\tau \simeq 1$ [sec], $l_0 \simeq 1$ [mm], and $\varepsilon \simeq \varepsilon_0 \simeq 10^{-11}$ [F/m], the required resistivity is obtained.

$$\rho \simeq 10^{11}\ [\Omega/\mathrm{m}], \quad \sigma \simeq 10^{14}\ [\Omega/\square]$$

This volume resistivity is higher than that of semiconductor and lower than that of common isolators. Some polymers such as nylon are in this domain but resistivity of nylon is dependent on temperature and humidity. Technique for its stabilization or new material should be developed.

Modifying surface resistivity by coating is a convenient way to control resistivity. It is easier to apply a coating than to produce a bulk film. However, the required surface resistivity for 1mm scale actuator is very high. It is higher than that of typical antistatic coating, which gives slight conductivity on isolative material to avoid collection of static charges. It is difficult to precisely make and maintain such high surface resistance because it is easily affected by environment such as humidity and contamination, and it easily gets mechanical damages. Soaking whole the actuator in dielectric liquid may help to control the environment to stabilize resistivity. It may also facilitate lubrication and heat transfer.

Control technique of resistivity is the key to development of the image charge actuator. Further study is required for this problem.

FABRICATION

We have fabricated the image charge actuator on PET (Poly Ethylene Terephthalate) films. Fig. 4 illustrates its cross section. To produce the slider, striped pattern of 0.42mm pitch is drawn at first using laser beam on a copper plated PET film. Then it is

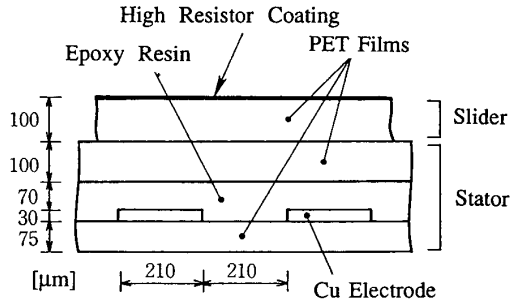


Fig. 4 Cross Section of Image Charge Actuator Fabricated On PET Film

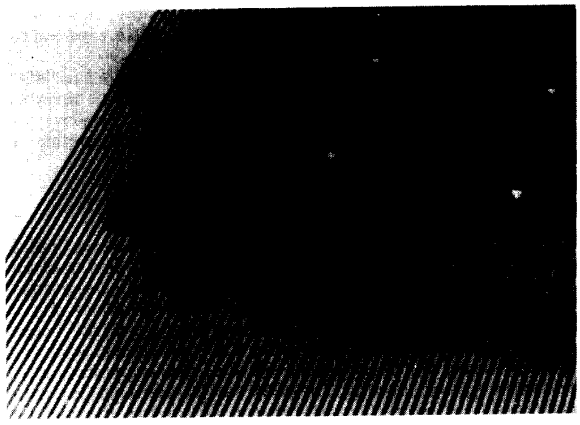


Fig. 5 Photo of Fabricated Film Actuator with 0.42mm Pitch Electrode

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etched to form the electrodes. The electrodes are covered with epoxy resin and a PET film. For the isolator layer of the slider, a PET film is used. A soft (high resistive) antistatic agent is coated on the film as the resistor layer. Fig. 5 shows the fabricated actuator. White dots seen on the slider are holes picked on the slider to prevent, by leading air, the stator and slider from sticking together.

When constructing the multi-layered actuator, film actuators are stacked in such a order as illustrated in Fig. 6. This configuration is preferable since, as well as the motive force, the normal repulsive force which reduces friction is produced on all the sliding surfaces. In this configuration, separation between layers is required to prevent the adjacent layers from affecting each other.

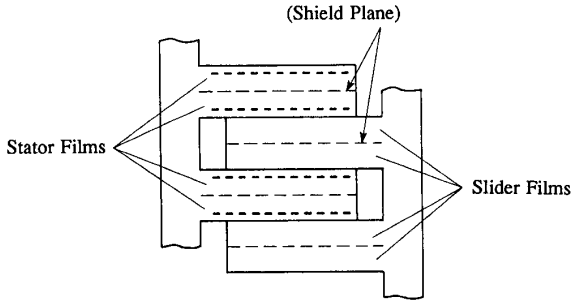


Fig. 6 Construction of Multi-Layered Actuator

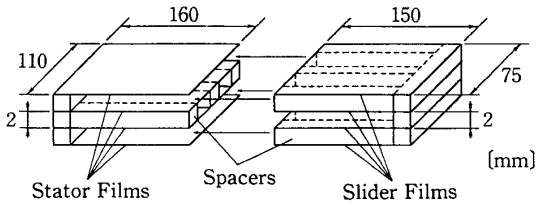


Fig. 7 Multi-Layered Actuator with Four Layers

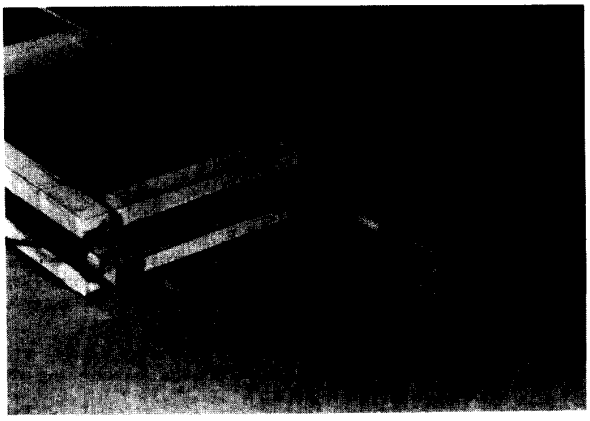


Fig. 8 Photo of Fabricated Multi-Layered Actuator

The required minimum separation is several times as much as pitch of electrodes. In addition to spatial separation, shield planes can be added, which completely reject electrical interference.

We have constructed a prototype of the multi-layered actuator by stacking four of the film actuators as shown in Fig. 7 and 8. Its effective interfacial area is 8cm×12.2cm per one layer and 390 cm^2 in total. Since about 2mm separation of layers is required this time, spacers of 2mm high are inserted. When the pitch is sufficiently reduced, spacers can be eliminated because the required separation becomes so small that thickness of the film itself can include it.

EXPERIMENTS

We applied ±800V on the film actuator through mechanical relays to drive it. It typically takes about 10 seconds to sufficiently induce charges on the slider which has no charges at first. The time constant varies with temperature, humidity, and aging of the resistor coating. It ranges from about 3 seconds to 30 seconds. Considering the relation between time constant and resistivity, surface resistivity of the coating is estimated as $10^{15}\ \Omega/\square$. After the charges are once induced, the time required for induction decreases to less than one second since the charge induced in the preceding induction process can be utilized repeatedly. The slider can move at minimum cycle time of 0.2 seconds when no load is applied, with ratio of voltage application time for charge induction and actuation stage selected to 9 : 1.

Fig. 9 shows a step movement of the slider observed by an optical instrument. The slider traverses distance of one pitch in 10 milliseconds. Since the charging time is too long compared with the traversing time, it can be noted that use of more appropriate material for the resistor layer will reduce initial charging time.

The multi-layered actuator has been operated with the same voltage applied. The spacers function effectively to avoid interference between the layers. Without the spacers, it occasionally reveals some improper operations such as rotation of films on their surfaces. We have measured the motive force produced by the multi-layered actuator. Some weights were used to exert a load on the actuator as shown in Fig. 10. Produced force is calculated as a sum of weight of the slider and the maximum load against which the slider can move. Force of this actuator stalls if its cycle time is decreased. In this experiment, it is fixed at 2 seconds. Measurement was done for the actuators with one, two, and four layers respectively. The force is approximately in proportion to the number of layers. This implies that the layers do not affect each other and their forces are effectively added. The actuator with four layers produces about 0.4N. Considering dimensions of the actuator, force density per unit interfacial area is 0.001N/cm^2 and that per unit volume is 0.004N/cm^3.

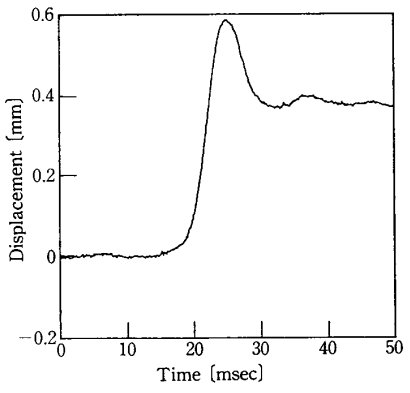


Fig. 9 Step Movement of Slider

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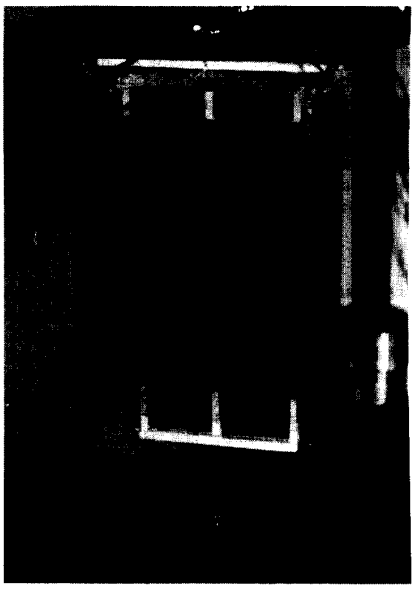


Fig. 10 Measurement of Force by Applying Weight

VARIATIONS

In this chapter we provide some variations of material, scale, and configuration of the image charge actuator.

A. Nylon Coating for Resistor Layer

As already mentioned, nylon has appropriate resistivity for the image charge actuator. We have tested performance of nylon coating. At first, nylon 6 pellets are dissolved in m-cresol. Then it is applied on a PET film to form a coating of about 5μm thick. The slider with nylon coating occasionally works properly. However, its resistivity easily varies with humidity and it gives no reliable result. The resistivity ranges from one hundredth to one hundred times of desirable value. Some stabilization technique should be developed or nylon coating can be used only in precisely controlled environment.

B. 0.1mm Pitch Film Actuator

A finer pitch actuator has been fabricated. Its stator has 0.1mm pitch electrodes on a 25μm thick polyimide film covered with polyimide. Reports on this actuator will appear in succeeding papers.

C. 3 DOF Actuation and 20μm Pitch Actuator

Since the slider of the image charge actuator is homogeneous in its extent, it can be placed at any arbitrary angle to the stator. This enables an actuator with three degrees of freedom to be realized. Fig. 11 shows an example of electrodes on the stator of a 3 DOF actuator with four groups of electrodes. Its slider can move to an arbitrary position and orientation in the plane, selecting the groups to be activated and determining their direction of thrust. Although this actuator does not fit for stacked structure, it will be useful for such applications as thin positioning tables and handling machines for thin materials. A 3 DOF actuator with 20μm pitch electrodes on alumina substrate have been designed. It will be fabricated before long.

D. Use of Photo Conductor

The resistor layer may be replaced by a photo conductor layer. Resistivity of the photo conductor is reduced by applying light while inducing charges. During the actuation stage, the light is cut off to prohibit the charges from running off. Advantages of the use of photo conductor include: (a) Performance of the actuator will be less affected by environment than when high resistor is used. When using photo conductor, the lower its resistivity under light is, in the shorter time charge induction can be done, and the higher its resistivity in the dark is, the more tightly the charges are fixed. Therefore, photo conductor with wider range of resistivity is preferable. If its resistivity under light is sufficiently low and that in the dark is sufficiently high, performance of the actuator will not change even if its resistivity somewhat varies. (b) Application of light reduces charge induction time. This improves time ratio of induction and actuation and the cycle time can be decreased.

The way to applying light is a problem in the use of photo conductor. When only one layer of the image charge actuator is used for applications such as positioning tables, light can be illuminated from above. In the multi-layered actuator, its films themselves may lead the light like optical fibers.

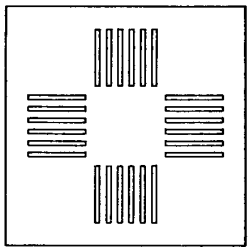


Fig. 11 Example of Electrode Pattern for 3 DOF Actuator

CONCLUSIONS

We have proposed the multi-layered electrostatic film actuator, which may be called electrostatic artificial muscle. To build that actuator, the image charge stepping actuator have been developed. As a prototype, we have fabricated an image charge actuator with 0.42mm pitch electrodes on PET films. Using four of this actuator, a multi-layered actuator have been constructed. Its performance have been examined by making it pull up some weights. It can operate under 0.4N load. The film actuators do not affect each other and the their force are effectively added. Though the obtained force density of $0.004N/cm^3$ is not yet satisfactory for ordinary scale machines such as robots, it is expected to be improved, reducing size of its elemental structure.

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